

# Two-Level DOPP: Conceptual Description (Current Version)

## Purpose

Two-Level DOPP is a budget-aware candidate selection strategy designed to reduce expensive full-solution evaluations. Instead of evaluating all candidate solutions directly, it first selects promising **regions** of the solution space, then selects representative **solutions** inside those regions.

In short:

- Region level decides **where to spend budget**.
- Solution level decides **which specific candidates to evaluate**.

This creates a structured exploration process that balances broad coverage and focused exploitation.

## Inputs and Data Objects

The current algorithm uses three aligned solution-level data views:

- **Solution feature matrix ( $X$ )**: numeric representation of each candidate solution.
- **Fitness vector ( $y$ )**: target quality value (lower is better in current usage).
- **Candidate keys**: stable IDs used to keep rows aligned across files and selections.

It also uses two proxy metrics per candidate:

- **Cut size**
- **Area imbalance**

These proxies are used only for region-level feature construction.

## Core Idea of the Two Levels

1. Cluster all solutions into balanced regions in feature space.
2. Build one feature vector per region.
3. Run D-optimal design on region features to pick informative regions (Round 1).
4. Inside each picked region, run D-optimal design again on solution features to pick informative solutions.
5. Use Round-1 region outcomes to train a region-level surrogate model.
6. Use the surrogate to pick additional unlabeled regions (Round 2).
7. Repeat inner solution selection and aggregate final evaluated candidates.

## Solution-Level Features

The algorithm treats solution-level features as a dense matrix where each row is one candidate.

How they are used:

- **Standardized solution features** are used for clustering and PCA representation.
- **Original solution features** are used for inner D-optimal selection inside each region.

No explicit solution feature-name tracking is used in the current workflow; the algorithm operates directly on numeric columns.

## Region-Level Features

Each region is converted into one feature vector by aggregating member solutions.

Current region feature groups are:

- Cut size statistics: min, max, mean, std
- Area imbalance statistics: min, max, mean, std
- Region size (number of member solutions)

- PCA-embedding statistics over region members:
- per-dimension mean
- per-dimension std

So each region feature vector is:

- 9 base dimensions from proxy stats + region size
- plus  $2 \times$  (number of PCA components)

After construction, QR-based rank cleanup removes linearly dependent region feature columns before region-level D-optimal design.

## Balanced Region Formation

Solutions are clustered into `K` regions after standardization.

Two balancing modes are supported:

- **\*\*Constrained balancing\*\*** (external constrained k-means package, if available)
- **\*\*Reassignment balancing\*\*** (standard k-means + iterative greedy rebalancing)

Goal: keep region sizes within a controlled range so the region-level process does not over-focus on very large clusters or under-represent smaller ones.

## Round 1 (Region-First Selection)

Round 1 runs D-optimal design on region features to get region weights. Top weighted regions are selected for expensive evaluation.

For each selected region:

- Run inner solution-level D-optimal design on that region's member solutions.
- Select a fraction of solutions by inner D-optimal weights.
- Evaluate selected solutions (via current precomputed fitness lookup flow).
- Record the best fitness found per region.

This produces labeled region outcomes used for surrogate training.

## Region Surrogate

A region-level regression model is fit to map region features to observed region quality (best fitness in that region from Round 1).

Current behavior:

- Uses Ridge when sample count is small relative to feature dimension.
- Uses linear regression otherwise.

The surrogate is used to score unlabeled regions for Round 2.

## Round 2 (Surrogate-Guided Selection)

Among regions not selected in Round 1:

- Predict region quality with the surrogate.
- Pick regions with best predicted quality (lowest predicted fitness).
- For those regions, run the same inner solution-level D-optimal selection and evaluation process.

This adds a second exploration wave guided by learned structure rather than only first-pass D-optimal weights.

## Final Outputs and Metrics

The pipeline aggregates both rounds and reports:

- All evaluated solution indices/keys
- Best solution found and its fitness
- Oracle call count (number of unique evaluated solutions)
- Per-round selected regions and per-region best outcomes
- Surrogate training metrics
- Round-2 prediction-vs-truth agreement (Kendall tau on eligible regions)
- Top-K coverage against the global true ranking

## **Practical Interpretation**

This implementation can be viewed as:

- **Phase A (representation and partitioning):** standardize, cluster, aggregate regions.
- **Phase B (structured sampling):** D-opt at region level, then D-opt within selected regions.
- **Phase C (adaptive refinement):** learn from Round 1 region outcomes and pick better Round 2 regions.

Compared with single-stage D-opt on all solutions, this two-level version emphasizes:

- Better budget allocation across diverse parts of the search space
- Improved sample efficiency under expensive evaluation constraints
- A clean handoff between unsupervised partitioning and supervised refinement